

Supply Chain & Inventory Optimization

Data-Driven Solutions for Replenishment and Cost Management

Problem Statement

- **Balancing Act:** The organization struggles to maintain the delicate balance between product availability and warehouse storage costs.
- **Cost Inflation:** Holding too much inventory leads to inflated warehousing costs and dead stock.
- **Lost Revenue:** Holding too little inventory results in immediate stockouts and unfulfilled demand.
- **Lack of Visibility:** Without real-time insights, manual tracking prevents efficient logistics planning.



The Challenge



Fluctuating Demand

Sales are highly seasonal, maintaining high volume in Q1/Q2 (~3,500 units/month) but dropping drastically by Q3 to below 1,300 units.



Replenishment Disconnect

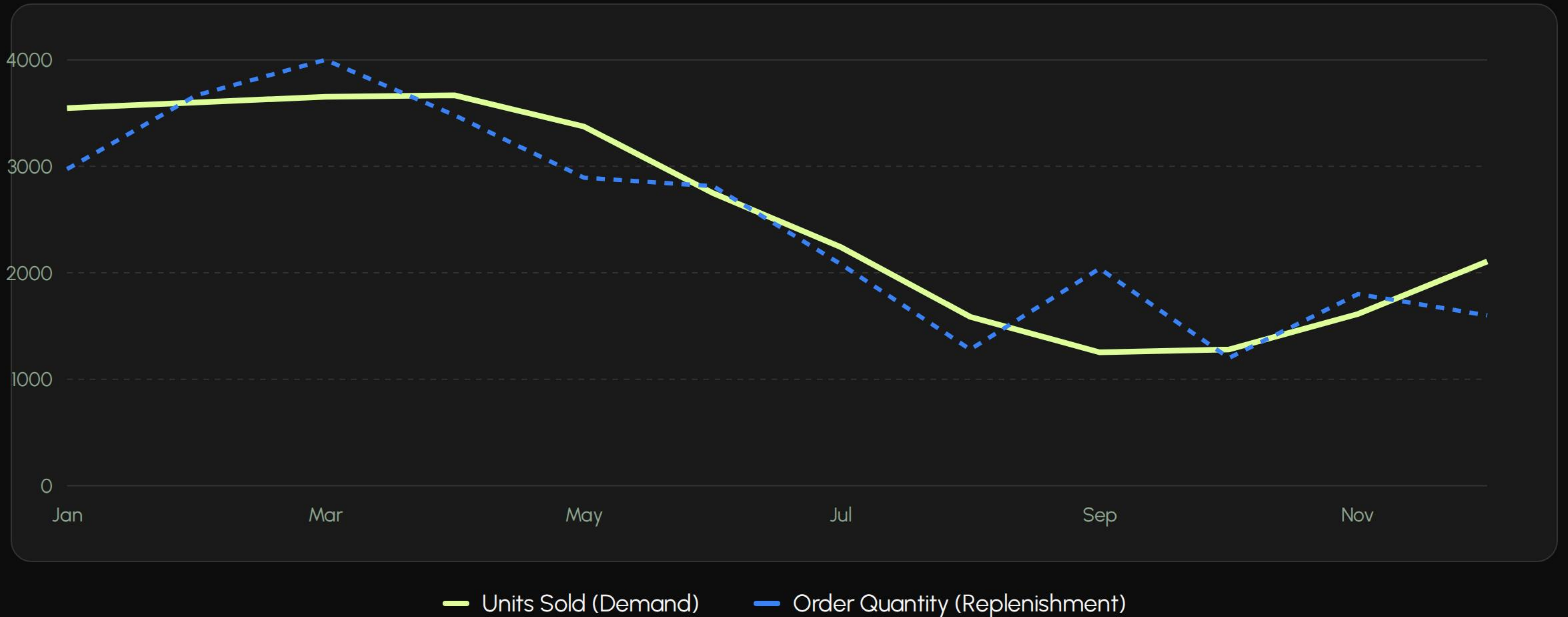
Order quantities do not align with demand drops. High volumes are still ordered in Q3, leading to severe inventory bloat by October.



Variable Lead Times

Supplier delivery times range unpredictably from 4 to 14 days, making it difficult to pinpoint the exact reorder point.

Seasonal Sales vs. Orders



Notice the dangerous disconnect in September (Sep): Sales drop to their lowest, yet replenishment orders spike, creating massive inventory bloat.

The Outcome

We developed a comprehensive, automated Excel-based optimization tool that transforms raw transactional data into actionable insights.

- **Live KPIs:** Instantly tracks metrics like Total Units Sold (30,730) and Total Inventory (764,143).
- **Dynamic Slicers:** Filter data by SKU, Warehouse, Supplier, and Region.
- **Multiple Views:** Three distinct dashboards tailored for different stakeholder needs.

back, Jay

12 months

24 hours

3

Overview

Education Sales

\$33K

Last Year: \$46K ↗ 14%

Commercial Sales

\$42K

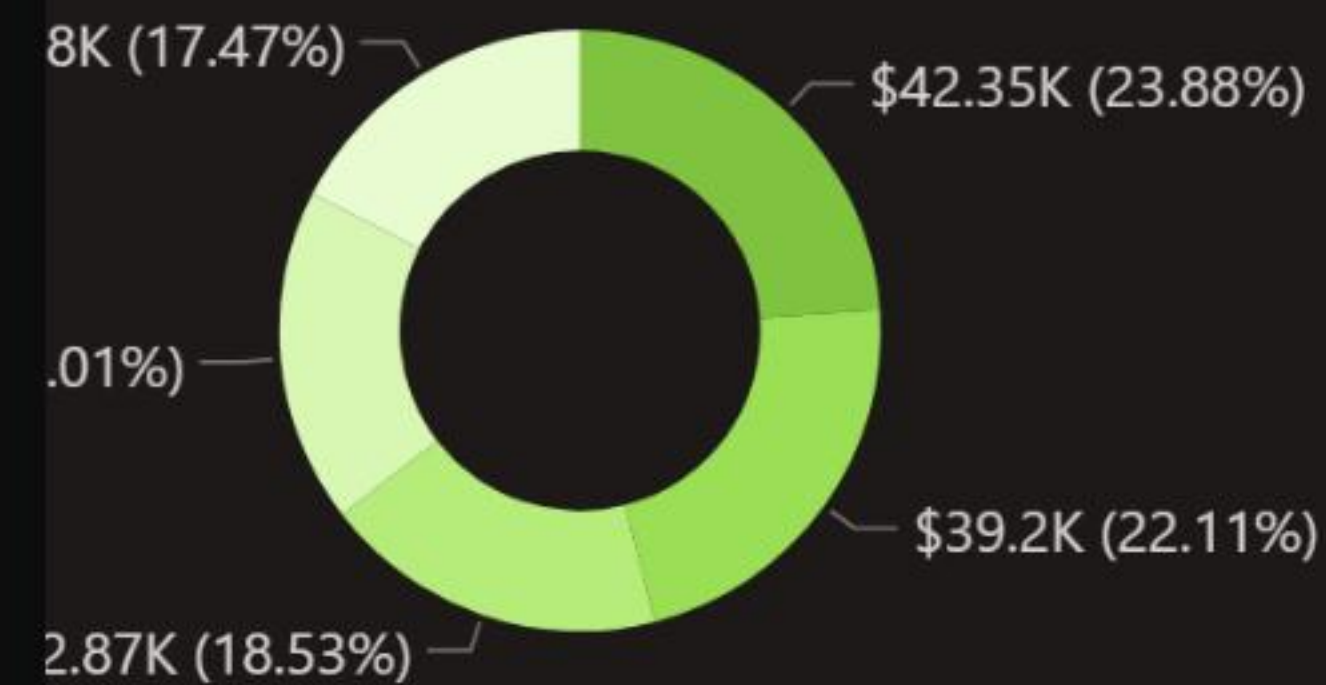
Last Year: \$62K ↗ 14%

Total Sales

\$177K

Last Year: \$256K

Area



Area

- Commercial
- Consumer
- Education
- Enterprise
- Federal



New Revenue

\$82K

LY: \$118K ↗ 14%



New O

\$72

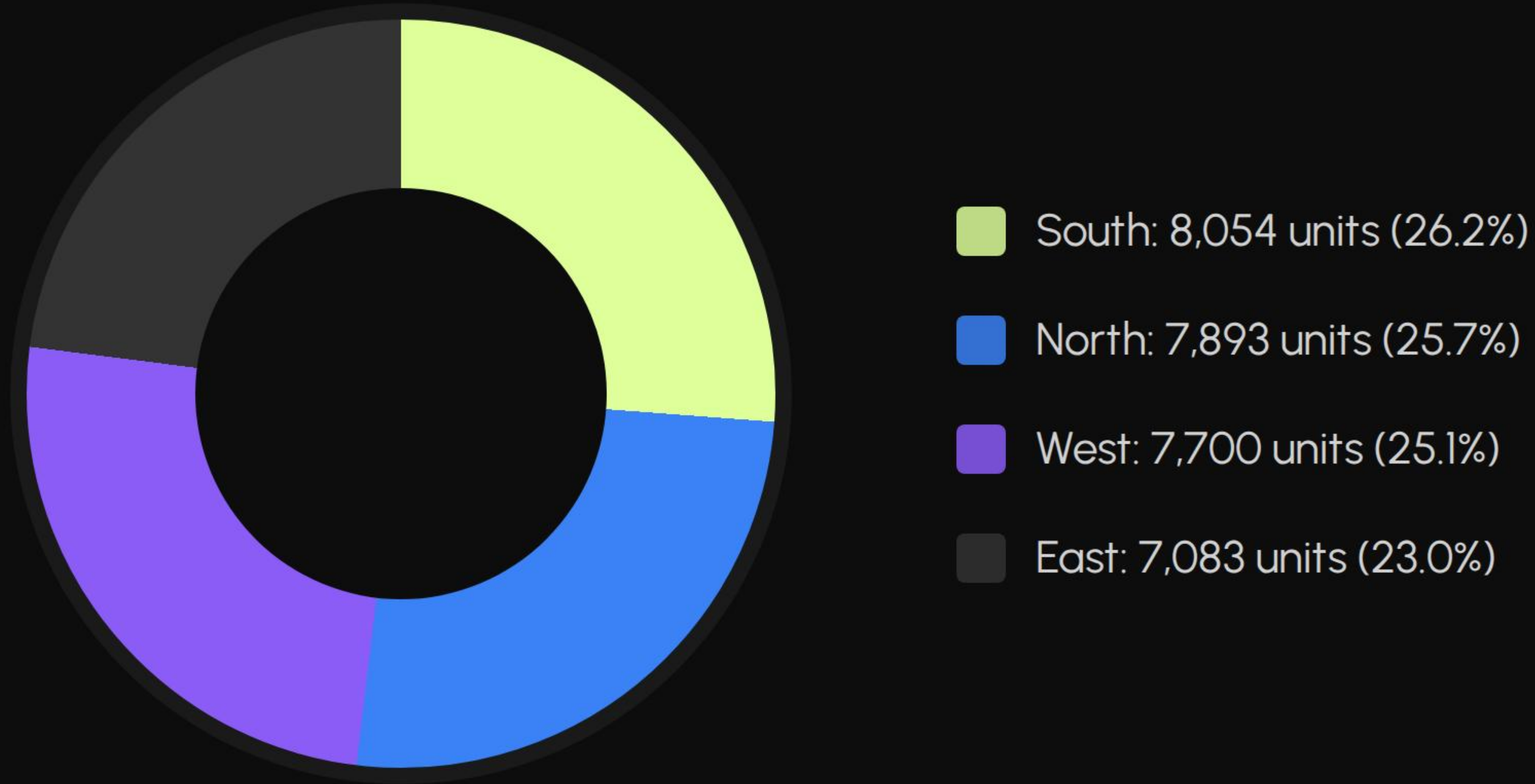
LY: \$10

Sum of Sales by Date



	Country Code	First Name	Last Name	City	Purchases
	IT	Daniel	Thomas	Rome	\$1,100
om	GB	David	Davis	London	\$1,000
	CA	Lisa	Thompson	Toronto	\$900

Regional Dominance



The South and North regions are the highest-performing sectors, representing over 50% of total sales. Inventory allocation must be prioritized to these critical regions.

Business Benefits



Cost Reduction

By aligning order quantities with the Q3/Q4 sales drop-off, the business eliminates end-of-year inventory bloat and slashes warehouse holding costs.



Optimized Replenishment

Dynamic reorder points (averaging 338 units) act as a necessary buffer to prevent stockouts during supplier lead times (up to 14 days).



Total Accessibility

The inclusion of the newly built "Instructions" sheet ensures that non-technical users and leadership can easily navigate the dashboards.

Questions?

Thank you for your time.

Image Sources



https://media.istockphoto.com/id/2173612915/photo/modern-distribution-center-with-smart-automation.jpg?s=612x612&w=0&k=20&c=i3XBdA6JvNFoaOoX9PGUJxdKyOt6b4wqnflOwa7_AFO=

Source: www.istockphoto.com



<https://i.redd.it/some-dark-theme-dashboards-i-made-v0-ooylkecwqk4d1.png?width=5124&format=png&auto=webp&s=d4ab685774c55bb3e6acd16280ddb2e048ebf26c>

Source: www.reddit.com
